Informatica® Metadata Command Center
October 2026

JDBC Sources

Informatica Metadata Command Center JDBC Sources
October 2026

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Preface

Read *JDBC Sources* to learn how to register and configure JDBC sources as catalog sources in Metadata Command Center. After you configure a catalog source, you extract metadata and then view the results in Data Governance and Catalog.

CHAPTER 1

Introduction to JDBC sources

You can use Metadata Command Center to extract metadata from a source system.

A source system is any system that contains data or metadata. For example, Apache Cassandra is a source system from which you can extract metadata through a JDBC catalog source. A catalog source is an object that represents and contains metadata from the source system.

Before you extract metadata from a source system, you first create and register a catalog source that represents the source system. Then you configure capabilities for the catalog source. A capability is a task that Metadata Command Center can perform, such as metadata extraction, lineage discovery, autonomous enrichment, data profiling, data classification, or glossary association.

When Metadata Command Center extracts metadata, Data Governance and Catalog displays the extracted metadata and its attributes as technical assets. You can then perform tasks such as analyzing the assets, viewing relationships, and creating links between those assets and their business context.

JDBC is a Java API to connect and interact with databases. You can use a JDBC catalog source to connect to specific source systems for which a JDBC driver is available.

You can connect to the following source systems with a JDBC catalog source:

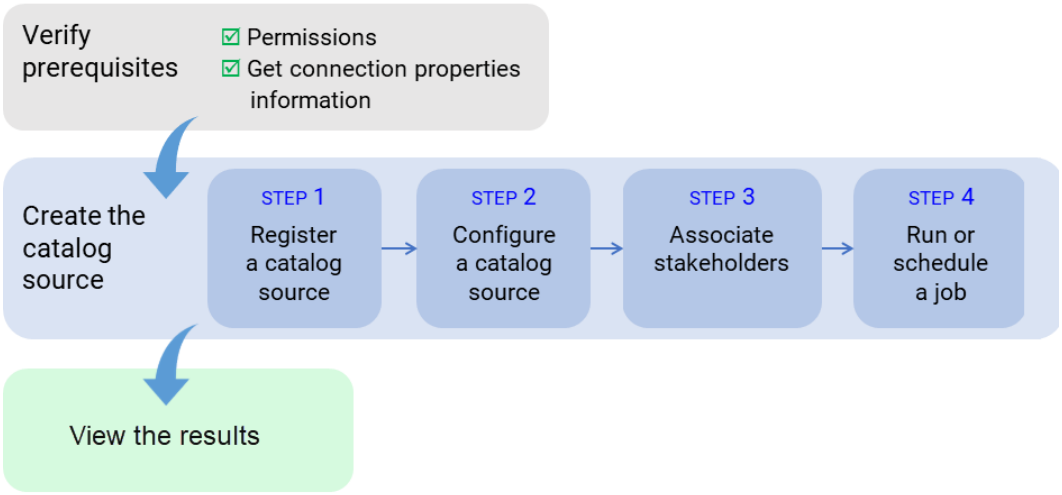
- Amazon Aurora MySQL
- Amazon DocumentDB
- Amazon DynamoDB
- Apache Cassandra
- Dremio
- IBM Db2
- Marketo
- Microsoft Azure Cosmos DB
- Microsoft Dynamics CRM
- MongoDB
- NetSuite
- SingleStore
- Sybase ASE (Adaptive Server Enterprise)
- Sybase IQ (Intelligent Query)
- Trino

Not all capabilities are available on all sources. See "About the JDBC catalog source" on page 7 for information about capabilities available on each source system.

Extraction and view process

To extract metadata from a source system, configure the catalog source and run the extraction job in Metadata Command Center. Then view the results in Data Governance and Catalog.

The following image shows the process to extract metadata from a source system:



After you verify prerequisites, perform the following tasks to extract metadata:

1. Register a catalog source. Create a catalog source object, select JDBC, and then select and test the connection.
2. Configure the catalog source. Specify the runtime environment and configure parameters for metadata extraction. Optionally, add filters to include or exclude source system assets from metadata extraction. You can also configure other capabilities such as lineage discovery, autonomous enrichment, data profiling and quality, data classification, or glossary association.
3. Optionally, associate stakeholders. Associate users with technical assets, giving the users permission to perform actions determined by their roles.
4. Run or schedule the catalog source job.

After you run the catalog source job, you view the results in Data Governance and Catalog.

About the JDBC catalog source

You can use the JDBC catalog source to extract metadata from a source system.

Although the catalog source includes multiple capabilities, not all capabilities are available on all source systems that you can connect to.

The following table lists source systems that you can connect to along with the metadata management capabilities that you can configure on each of them:

Source System	Data Profiling and Data Quality Automation	Data Observability	Data Classification	Glossary Association
Amazon Aurora MySQL	Yes	Yes	Yes	Yes
Amazon DocumentDB	No	No	No	No
Amazon DynamoDB	No	No	Yes	Yes
Apache Cassandra	No	No	Yes	Yes
Dremio	No	No	No	No
IBM Db2	Yes	Yes	Yes	Yes
Marketo	No	No	Yes	Yes
Microsoft Azure Cosmos DB	No	No	Yes	Yes
Microsoft Dynamics CRM	No	No	Yes	Yes
MongoDB	No	No	Yes	Yes
NetSuite	No	No	Yes	Yes
SingleStore	No	No	Yes	Yes
Sybase ASE (Adaptive Server Enterprise)	No	No	Yes	Yes
Sybase IQ (Intelligent Query)	No	No	Yes	Yes
Trino	Yes	No	No	No

You can configure the following capabilities:

Data Profiling and Data Quality Automation

- **Data Profiling**. Assesses source metadata and analyzes the collected statistics to discover content and structure, such as value distribution, patterns, and data types.
- **Data Quality**. Measures the reliability of the data and enables data usage.

Data Observability

Identifies anomalies in the characteristics of the data that you profile.

Data Classification

Data classification is the process of identifying and organizing data into relevant categories based on the functional meaning of the data. Classifying data can help your organization manage risks, compliance, and data security.

Glossary Association

You can associate terms that are in the glossary with technical assets to provide user-friendly business names to technical assets. Glossary Association automatically associates glossary terms with technical assets or recommends glossary terms that you can manually associate with technical assets in Data Governance and Catalog.

Autonomous Enrichment

Automatically propagates classification, description, business name, and glossary enrichments from an asset to related assets based on the lineage between assets. Autonomous enrichment reduces manual curation effort by extending the enrichments that you apply to one asset across assets that are connected to it through lineage.

Extracted metadata

You can extract specific metadata from source systems that you connect to with a JDBC catalog source.

Objects extracted

Metadata Command Center extracts the following metadata from source systems that you connect to with the JDBC catalog source:

- Database
- Schema
- View
- Table
- Column
- View Column

Data profiling for Amazon Aurora MySQL objects

Configure data profiling to run profiles on objects that you extract from an Amazon Aurora MySQL source.

You can run profiles on the following object types:

- Char
- Date
- Decimal
- Double
- Float
- Integer
- Smallint
- Bigint
- Time
- Text
- Mediumtext
- Tinytext
- Longtext
- Varchar

- Character varying

Data profiling for IBM Db2 objects

Configure data profiling to run profiles on objects that you extract from an IBM Db2 source.

You can run profiles on the following object types:

- Bigint
- Char(L)
- Date
- Decimal(P,S)
- Float
- Graphic
- Integer
- Numeric(P,S)
- Smallint
- Time
- Timestamp
- Varchar
- Vargraphic

Data profiling for Trino objects

Configure data profiling to run profiles on objects that you extract from a Trino source.

You can run profiles on the following object types:

- Bigint
- Boolean
- Char
- Char (N)
- Date
- Decimal
- Decimal (P,S)
- Double
- Integer
- Number
- Real
- Smallint
- Time
- Time (P)
- Time with Time Zone
- Time (P) with Time Zone

- Timestamp
- Timestamp (P)
- Timestamp with Time Zone
- Timestamp (P) with Time Zone
- Tinyint
- Varchar
- Varchar (N)

Compatible connectors

Before you configure a catalog source, you must connect to the JDBC source system.

Use the JDBC V2 connector to connect to the JDBC source system.

For information about configuring a connection, see *Connections* in the Administrator help.

CHAPTER 2

Before you begin

Before you create a catalog source, ensure that you have the information required to connect to the source system.

Perform the following tasks:

- Assign the required permissions.
- Install the latest Type 4 JDBC driver version on the Secure Agent machine.
- Configure a JDBC V2 connection to connect to the source system in Administrator.

Verify permissions

To extract metadata and to configure other capabilities that a catalog source might include, you need account access and permissions on the source system. The permissions required might vary depending on the capability.

Permissions for metadata extraction

Ensure that you have the required permissions to perform metadata extraction.

Create a user account for the Informatica user to access the JDBC source system. Grant read permission to the new user account.

Permissions to run data profiles

You can run profiles with the permissions required on the source system to perform metadata extraction.

Permissions to run data classification

You can perform data classification with the permissions required on the source system to perform metadata extraction.

Permissions to run glossary association

You can perform glossary association with the permissions required on the source system to perform metadata extraction.

Install the latest Type 4 JDBC driver on the Secure Agent machine

To connect to the JDBC source system, the Secure Agent machine must have JDBC drivers compatible with the source system.

Perform the following steps to install the JDBC drivers on the Secure Agent machine:

1. Navigate to the following directory on the Secure Agent machine:
   `<Secure Agent installation directory>/ext/connectors/thirdparty/informatica.jdbc_v2/common`
2. Download the latest Type 4 JDBC driver version that your database supports from the third-party vendor site.
   For example, if you want to use the JDBC V2 connector to connect to Amazon Aurora MySQL, download the Amazon Aurora MySQL driver. Amazon Aurora MySQL driver version 8.0.27 works with the JDBC V2 connector.

   You can add multiple drivers for different source systems.
3. Restart the Secure Agent.

Create a connection

Before you configure the JDBC catalog source, create a connection object in Administrator.

1. In Administrator, select **Connections**.
2. Click **Add Connection**.
3. Search for and select **JDBC V2** and then click **Next**.
4. Enter the following connection details:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.

Property	Description
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. When you enable the secret vault in the connection, you can select the credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.
Runtime Environment	The name of the runtime environment where you want to run tasks. Select a Secure Agent.
JDBC Driver Class Name	Name of the JDBC driver class. For example, to connect to Apache Cassandra, specify the following driver class name: cdata.jdbc.cassandra.CassandraDriver For more information about which driver class to use with specific databases, see the corresponding third-party vendor documentation.
Connection String	Connection string to connect to the database. Use the following format to specify the connection string: `jdbc:<subprotocol>:<subname>` For example, the connection string for the Apache Cassandra database type is `jdbc:cassandra://<host>:<port>[/keyspacename].` For more information about the connection string to use with specific drivers, see the corresponding third-party vendor documentation.
User Name	The user name to connect to the database.
Password	The password to connect to the database.

5. In the **Advanced Settings** section, you can choose to enter sensitive information such as the client ID, client secret, or refresh token in the following property:

Property	Description
Additional Security Properties	Masks sensitive and confidential data of the connection string that you don't want to display in the session log. Specify the part of the connection string that you want to mask. When you create a connection, the string you enter in this field appends to the string that you specified in the **Connection String** field.

6. Click **Test** to test your connection to the source system.
7. Click **Save**.

CHAPTER 3

Create catalog sources in Metadata Command Center

Use Metadata Command Center to configure a catalog source for JDBC and extract metadata.

When you configure a catalog source, you define the source system where you want to extract metadata from. Configure filters to include or exclude source system metadata before you run the job. Optionally, configure other capabilities, such as data profiling and quality, data classification, relationship discovery, autonomous enrichment, and glossary association. To provide stakeholders access to technical assets, you can assign access through stakeholder roles. You can also associate technical assets extracted from the catalog source to asset groups.

Step 1. Register a catalog source

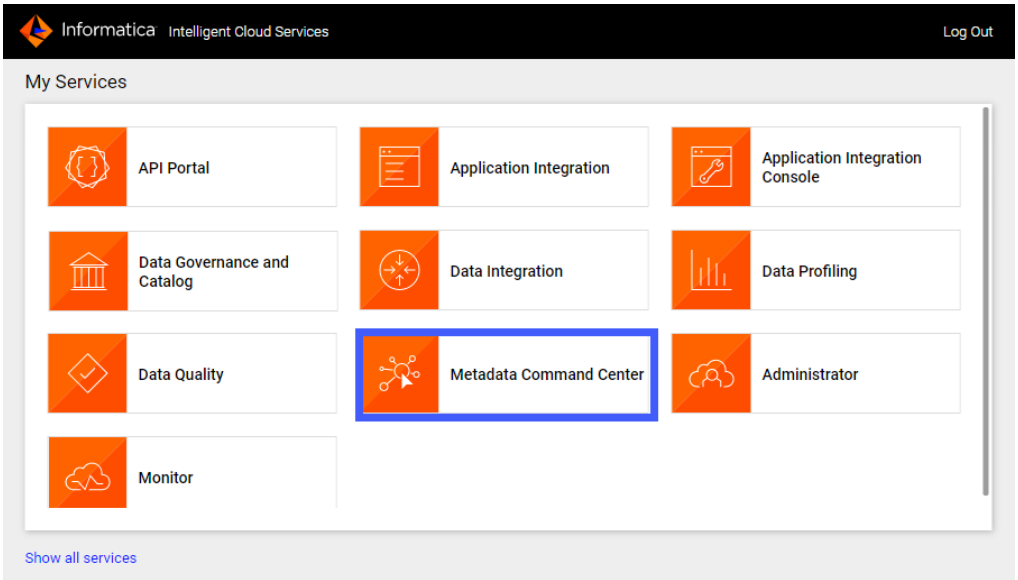
When you register a catalog source, provide general information and connection values.

1. Log in to Intelligent Data Management Cloud.

   The **My Services** page appears.

2. Click **Metadata Command Center**.

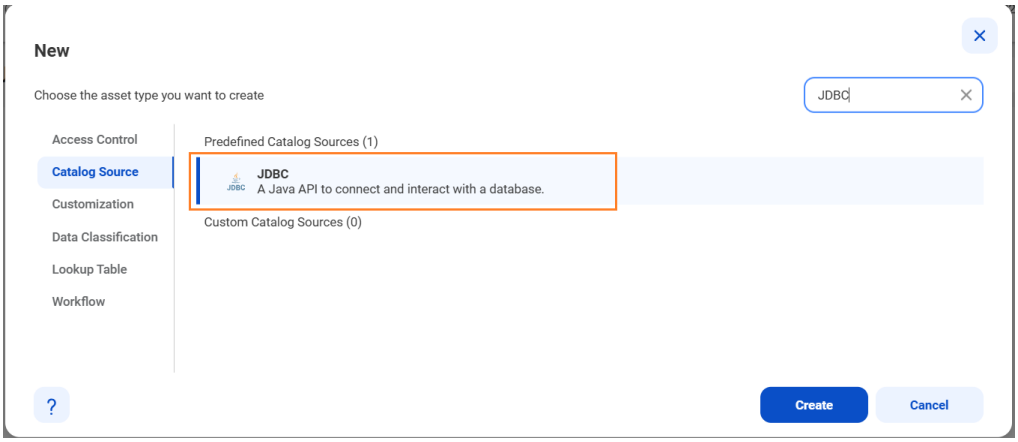
   The **Metadata Command Center** box appears on the **My Services** page.



The Metadata Command Center home page appears.

3. Click **New**.
4. Select **Catalog Source** from the list of asset types.
5. Select JDBC from the list of catalog source types.

   The New Catalog Source dialog box opens with the JDBC catalog source type.



6. Click **Create**.

   The New Catalog Source page opens.

7. In the **General Information** section, enter a name and an optional description for the catalog source.

   Note: You can rename a catalog source after you create it, but to apply the change to all associated objects you must rerun the metadata extraction job.

   After you save the catalog source, you can update the description in Metadata Command Center and Data Governance and Catalog. The update appears only in the service in which you update it.

8. In the **Connection Information** area, select the connection that you created in Administrator.

   Note: To create or edit a catalog source, you need permissions on the connection to the source system. Select a connection that you have access to, or ask the administrator to grant the necessary permissions to the connection that you want to use.

9. Click **Connection Properties** to expand and view the connection properties for the selected connection.
10. Click **Test Connection** to test your connection to the source system.
11. Click **Next**.

The **Configuration** page appears.

Step 2. Configure capabilities

When you configure the JDBC catalog source, you define the settings for the metadata extraction capability and other optional capabilities.

The metadata extraction capability extracts source metadata from external source systems. You can also configure other capabilities that the catalog source includes.

You can save the catalog source configuration at any point after you enter the connection information. After you save the catalog source, you can choose to run the catalog source job. To run the job once, click **Run**. To run metadata extraction and other capabilities on a recurring schedule, configure schedules on the **Schedule** tab.

Configure metadata extraction

When you configure the JDBC catalog source, you choose a runtime environment and enter configuration parameters for metadata extraction.

1. In the **Connection and Runtime** area, choose a serverless runtime environment or the Secure Agent group where you want to run catalog source jobs.

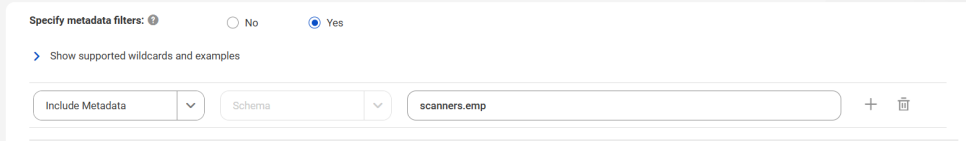
   Note: Serverless runtime environment options are available if the catalog source works with a serverless runtime environment.
2. Choose to retain, delete, or deprecate objects that are deleted from the source system in the catalog with the **Metadata Change Option**.
   - **Retain**. Retains objects that are deleted from the source system in the catalog. If you update or add a filter, the catalog retains objects extracted from the previous job and extracts additional objects that match the current filter. Objects deleted from the source system are not deleted from the catalog. Enrichments added on deleted objects and relationships are retained.
   - **Delete**. Deletes metadata from the catalog based on objects deleted from the source system and changes you make to the filter. Enrichments added on deleted objects and relationships are also permanently lost. Objects renamed in the source system are removed and recreated in the catalog.
   - **Deprecate**. When objects are deleted from the source system, the catalog changes the lifecycle of the imported objects to Obsolete. The lifecycle of objects also changes to Obsolete when you add exclusion filters or reduce the scope so that objects extracted in earlier jobs are no longer included. For example, the first job includes the sales schema. The second job also includes the sales schema, but excludes the sales.transactions table. In this case, the second job marks the sales.transactions object as Obsolete, even though the first job includes it. This does not impact enrichments applied to deprecated objects and relationships. Objects renamed in the source system are removed and recreated in the catalog. When you run the catalog source job again for other capabilities such as data classification, relationship discovery, or glossary association, the job doesn't consider obsolete objects. Obsolete objects remain in the catalog until they are purged when you run a **Purge Obsolete Objects** job on the **Explore** page.

   Note: You can also change the configured metadata change option when you run a catalog source.

3. In the **Filters** area, define one or more filter conditions to apply for metadata extraction.
   a. Select **Yes** to view filter options.
   b. From the Include or Exclude metadata list, choose to include or exclude metadata based on the filter parameters.
   c. Enter the filter values.

      Filters can contain an asterisk wildcard to represent multiple characters.

      The value appears in the filter value field.



   d. To define an additional filter with an OR condition, click the **Add** icon.
4. Optional. In the **Configuration Parameters** area, enter additional settings.

   The following table describes the property that you enter for additional settings:

   Note: The **Additional Settings** section appears when you click **Show Advanced**.

Property	Description
Expert Parameters	Enter additional configuration options to be passed at runtime. Required if you need to troubleshoot the catalog source job. **Caution:** Use expert parameters when it is recommended by Informatica Global Customer Support.

5. Configure additional capabilities for the catalog source by clicking on the tabs.

Configure data profiling and quality

Enable the data profiling capability to evaluate the quality of metadata extracted from the JDBC source system.

1. Click the **Data Profiling and Quality** tab.
2. Expand **Data Profiling** and select **Enable Data Profiling**.

   Note: Ensure that you have permissions on all the staging connections that you use in your data profiling configuration. You can't run the job if you don't have permissions on the connections that you use. Select connections that you have access to, or ask the administrator to grant the necessary permissions on the connections that you want to use.
3. In the **Connection and Runtime** area, choose the Secure Agent group where you want to run catalog source jobs.
4. Optionally, in the **Filters** area, specify additional filters in addition to metadata filters:
   a. Select **Yes** to view filter options.
   b. From the Include or Exclude list, choose to include or exclude metadata based on the filter parameters.
   c. Enter the path to the object as the filter value.

      Filters can contain an asterisk wildcard to represent multiple characters.

Examples:

- You extracted metadata from all tables and views from a schema and now you want to profile a specific table. Enter the schema name followed by the table name in the input field. For example, `Schema_Name.TABLE_NAME`
- You extracted metadata from multiple schemas and now you want to run a profile on all objects in a specific schema. Enter the schema name in the input field.

To include or exclude multiple objects, click the **Add** icon to add filters with the OR condition.

5. In the **Parameters** area, configure the parameters.

   The following table describes the parameters that you can enter:

Parameter	Description
Modes of Run	Determine the type of data that you want the data profiling task to collect. Choose one of the following options: • Keep signatures only. Collects only aggregate information such as data types, average, standard deviation, and patterns. • Keep signatures and values. Collects both signatures and data values.
Profiling Scope	Determine whether you want to run data profiling only on the changes made to the source system or on the entire source system. Choose one of the following options: • Incremental. Includes only source metadata that is changed or updated since the last profile run. • Full. Includes the entire metadata that is extracted based on the filters applied for extraction.
Sampling Type	Determine the sample rows on which you want to run the data profiling task. Choose one of the following options: • All Rows. Runs data profiling on all rows in the metadata. • Custom Query. Provide a custom SQL query to select sample rows to run the data profiling task. In the **Sampling Query** field, enter the custom SQL query to choose sample rows on which you want to run the data profiling task.
Maximum Precision of String Fields	The maximum precision value for profiles on string data type.
Text Qualifier	The character that defines string boundaries. If you select a quote character, profiling ignores delimiters within the quotes. Select a qualifier from the list. **Note:** Default is Double Quote.

6. Expand **Data Quality** and select **Enable Data Quality**.

   Note: You can click **Use Data Profiling Parameters** to use the same parameters as in the **Data Profiling** section.

   Note: Ensure that you have permissions on all the staging and flat file connections that you use in your data quality configuration. You can't run the job if you don't have permissions on the connections that you use. Select connections that you have access to, or ask the administrator to grant the necessary permissions on the connections that you want to use.

7. In the **Connection and Runtime** area, choose the Secure Agent group where you want to run catalog source jobs.
8. In the **Parameters** area, configure the parameters.

   The following table describes the properties that you can enter:

Parameter	Description
Data Quality Rule Automation	Enable the option to automatically create or update rule occurrences for data elements in the catalog source. Choose one of the following options: • Apply on Data Elements linked with Business Dataset. Creates rule occurrences for all data elements that are linked with business data sets in the catalog source. • Apply on all Data Elements. Creates rule occurrences for all data elements in the catalog source.
Data Quality Remediation	Enable the option to specify a flat file connection to store the list of failed rows so that users can remediate poor data quality scores. Choose one of the following options: • **No**. Doesn't enable the **Create Data Quality Failure Ticket** option. • **Yes**. Shows a list of flat file connections where you write failed rows to customer-managed locations.
Create Data Quality Failure Ticket	Required if you enable data quality remediation. Specify whether you want to create data quality failure tickets for poor data quality scores based on the threshold defined for the rule occurrence in Data Governance and Catalog. Choose one of the following options: • **No**. Doesn't automatically create data quality failure tickets when the data quality scores are poor. • **Yes**. Automatically creates data quality failure tickets based on the data quality threshold values you define in Data Governance and Catalog, and notifies you when a data quality score is below the threshold. **Note:** You must configure a workflow event for the data quality failure and enable the event in Metadata Command Center.
Cache Result	Specify how you want to preview rule occurrence results. Select **Agent Cache** if you want to generate a cache file in the runtime environment and to preview the cached results faster in subsequent data preview runs. The results are cached for seven days by default after the first run in the runtime environment. Select **No Cache** if you don't want to cache the preview results and view the live results.
Run Rule Occurrence Frequency	Specify whether you want to run data quality rules based on the frequency defined for the rule occurrence in Data Governance and Catalog.
Sampling Type	Determine the sample rows on which you want to run the data quality task. Choose one of the following options: • All Rows. Runs data quality on all rows in the metadata. • Custom Query. Provide a custom SQL clause to select sample rows to run the data quality task. In the **Sampling Query** field, enter the custom SQL query to choose sample rows on which you want to run the data quality task.

Parameter	Description
Maximum Precision of String Fields	The maximum precision value for profiles on string data type.
Text Qualifier	The character that defines string boundaries. If you select a quote character, profiling ignores delimiters within the quotes. Select a qualifier from the list. **Note:** Default is Double Quote.

Configure data observability

Enable the data observability capability to receive anomaly notifications for metadata extracted from the source system.

To include profiling anomalies, enable data profiling for the catalog source. For accurate results, set the **Profiling Scope** option to **Full** in the **Data Profiling and Quality** configuration.

1. Click the **Data Observability** tab.
2. Select **Enable Data Observability**.
3. In the **Parameters** area, enter a value to specify the minimum number of catalog source job runs required to start detecting anomalies. For example, if you enter a value of 5, then the job starts to detect anomalies from the sixth run.

   Note: This option does not apply to anomalies that require a minimum of two runs to generate.
4. Specify the maximum number of anomaly events to generate for each catalog source run. Default is 1000.
5. Optionally, in the **Asset Filters** area, enable filters to run data observability on specific assets.

   Specify data profiling filters if you want to override the data profiling filter and extract profiling anomalies for assets based on a different filter.
6. To add a data profiling filter, perform the following steps:
   a. Select **Yes** to view filter options.
   b. From the **Include/Exclude** list, choose to include or exclude metadata.
   c. From the **Specify data profiling filters** list, select an object type.
   d. Enter a value to specify the object location.

      For example, you ran a profile on multiple schemas, and now you want to generate profiling anomalies on a specific schema. Enter the schema name in the input field.
   e. To include or exclude multiple objects, click the **Add** icon to add filters with the OR condition.
7. Optionally, in the **Metric Filter** area, specify filters to generate events based on specific metrics. If you don't include a filter, the job detects both profiling and other anomalies applicable to the catalog source.
   a. Select **Filter conditions**.
   b. From the **Include Metric** list, choose to include or exclude a metric.
   c. From the **All Metrics** list, select a metric for which you want to generate events.
   d. From the **Sensitivity** list, select the sensitivity of the anomaly.
   e. From the **Select detection rules** list, select rules that you apply to detect anomalies.

The following table describes the detection rules that you can select:

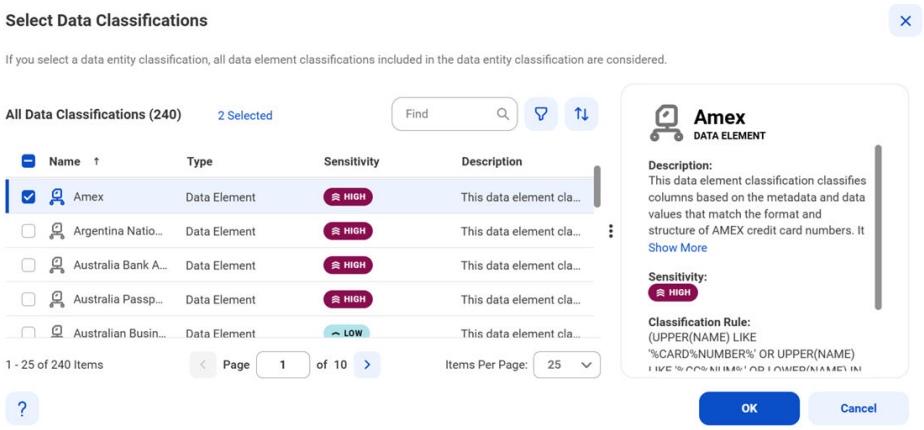
Option	Description
Static Data	Detects the following anomalies: - Percentage variation. If percentage values other than 0% or 100% remain constant for three or more profile runs, any subsequent change generates a Percentage Variation anomaly. - Count variation. A change in a value that is usually constant indicates that there is an anomaly.
100% or 0% Change Detection	Detects the following anomalies: - Drop from maximum. A drop from 100% to a lower value generates a Drop from Maximum anomaly. - Surge from minimum. A spike from 0% to any higher value generates a Surge from Minimum anomaly.
Standard Deviation	Detects the following anomalies: - Drop in transition. A drop from a value that has been decreasing at a constant rate to a significantly lower value generates a Drop in Transition anomaly. - Surge in transition. A spike from a value that has been increasing at a constant rate to a significantly higher value generates a Surge in Transition anomaly. - Deviation. If the percentage values of a record change with each profile run, the algorithm calculates an expected range based on the changes observed in each profile run. A new value that falls outside of this expected range generates a Deviation anomaly.
Breaking Trends	Detect the following types of count-based anomalies: - Drop. A drop in a value that has been increasing to a lower value generates a Drop anomaly. - Surge. A spike in a value that has been decreasing to a higher value generates a Surge anomaly.

Configure data classification

Enable the data classification capability to identify and organize data into relevant categories based on the functional meaning of the data.

1. Click the **Data Classification** tab.
2. Select **Enable Data Classification**.
3. Choose one or both of the following options:
   - **Generated Data Classifications**. CLAIRE automatically generates data classifications for the data elements.

- **Data Classification Rules**. Choose from predefined or custom data classifications.
  1. Click **Add Data Classification**. The **Select Data Classifications** dialog box appears.



  2. Select the data classifications that you want to use.
  3. Click **OK**.

Configure glossary associations

Enable the glossary association capability to associate glossary terms with technical assets, or to get recommendations for glossary terms that you can manually associate with technical assets in Data Governance and Catalog.

Metadata Command Center considers all published business terms in the glossary while making recommendations to associate your technical assets.

1. Click the **Glossary Association** tab.
2. Select **Enable Glossary Association**.
3. Select **Enable auto-acceptance** to automatically accept glossary association recommendations.
4. Specify the **Confidence Score Threshold for Auto-Acceptance** to set a threshold limit based on which the glossary association capability automatically accepts the recommended glossary terms.

   Note: Specify a percentage from 80 to 100. If the score is higher than the specified limit, the glossary association capability automatically assigns a matching glossary term to the data element.
5. Select **Enable Below-threshold Recommendations** to receive glossary association recommendations below the auto-acceptance threshold. If you enable auto-acceptance, you can enable below-threshold recommendations to receive glossary recommendations below the auto-acceptance threshold.
6. Specify the **Confidence Score Threshold for Recommendations** to set a threshold based on which the glossary association capability makes recommendations

   If you enable auto-acceptance, specify a percentage from 80 to the selected auto-acceptance threshold. You can accept or reject the recommended glossary terms that fall within this range in Data Governance and Catalog.

   If you disable auto-acceptance, specify a percentage from 80 to 100 inclusive.
7. Choose to automatically assign business names and descriptions to technical assets. You can then choose to retain existing assignments and only assign business names and descriptions to assets that don't have assignments, or allow overwrite of existing assignments.

   By default, existing assignments are retained.

8. Optional. Choose to ignore specific parts of data elements when making recommendations. Select **Yes** and enter prefix and suffix keyword values as needed.

   Click **Select** to enter a keyword. You can enter multiple unique prefix and suffix keywords. Keyword values are case insensitive.

9. Optional. Choose specific top-level business glossary assets to associate with technical assets. Selecting a top-level asset selects its child assets as well. Select **Top-level Glossary Assets** and specify the assets on the **Select Assets** page.
10. Optional. Choose to use abbreviations and synonym definitions from lookup tables for accurate glossary association. Select **Yes** to enable, and then click **Select** to upload a lookup table.
11. Click **Next**.

    The **Associations** page appears.

Configure automation

Enable lineage-based propagation to receive enrichments propagated from catalog sources connected through lineage.

Enable Lineage-based Propagation on the Autonomous Enrichment tab of the Configure page before you configure this catalog source to receive enrichments. For more information, see *Administration*.

With the lineage-based propagation settings on this tab, you configure how this catalog source receives enrichments propagated from connected catalog sources. Enrichments propagate only to assets that don't already have enrichments.

1. Click the **Automation** tab.
2. In the **Autonomous Enrichment** section, select **Enable Lineage-based Propagation**.

   The lineage-based propagation settings appear.
3. Select the enrichment types to receive from connected catalog sources.

   You can select Classification, Description, Business Name, and Glossary Term.
4. Choose how to apply received enrichments.
   - Select **Auto-accepted** to apply received enrichments automatically.
   - Select **Recommended** to list them as suggestions that you can curate in Data Governance and Catalog.
5. Choose whether to receive enrichments only from certified assets.

   Assets are certified in Data Governance and Catalog. This option is selected by default. If you clear it, this catalog source also receives enrichments from assets that are not certified, which might be less reliable.
6. Choose where this catalog source receives enrichments from.
   - Select **Catalog Source** to receive enrichments from a specific catalog source.
   - Select **Nearest Neighbor** to receive enrichments from the closest connected catalog source based on lineage.
7. If you select **Catalog Source**, select a catalog source from the list.

   Ensure that the catalog source you select is within 15 lineage hops. Enrichments don't propagate from a catalog source beyond that distance.

8. If you select **Nearest Neighbor**, configure the lineage settings:
    a. Set **Lineage Direction** to search upstream, downstream, or in both directions.

       Upstream catalog sources feed data into this catalog source. Downstream catalog sources receive data from this catalog source. Select both directions to search upstream and downstream at the same time.
    b. Set **Number of Hops** to define how far along lineage to search.

       The search stops at the first connected catalog source that has enrichments, up to a maximum of 15 hops.
9. Click **Save**.

When you save, the catalog source begins receiving enrichments created from that point onward. Saving doesn't retrieve enrichments created prior to this date and time. To receive such enrichments, run the catalog source job manually and include the Autonomous Enrichment capability.

Step 3. Associate stakeholders and asset groups

Associate users or user groups within a stakeholder role as stakeholders for technical assets in Data Governance and Catalog. Also, you can choose to assign technical assets extracted from the catalog source to asset groups. You can then use access policies to control permissions on assets that are assigned to asset groups.

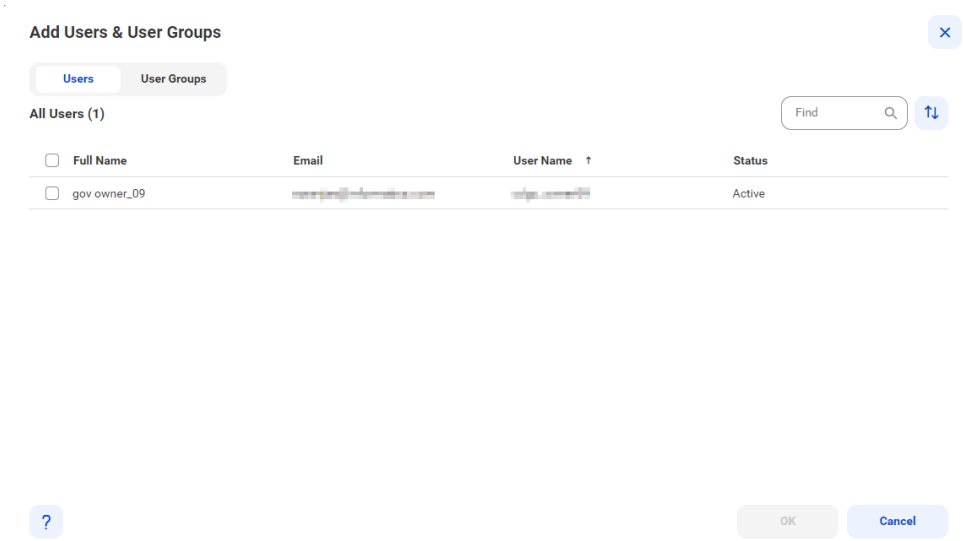
Verify that the administrator assigned users and user groups to the stakeholder role that you want to associate with technical assets.

1. To associate users or user groups as stakeholders with technical assets extracted from the catalog source, perform the following steps:
    a. On the **Associations** page, click **Stakeholders**.
    b. Select **Assign Stakeholders**.
    c. Select **Propagate to Child Assets** to propagate stakeholders of a parent asset to child assets without stakeholders.
    d. Select **Append Stakeholders** to propagate stakeholders of a parent asset to all child assets including assets with stakeholders assigned.

       Note: To enable **Append Stakeholders**, you must enable **Propagate to Child Assets**.

       You can also override the propagation settings from the catalog source page in Data Governance and Catalog. For more information about stakeholder propagation, see *Stakeholder Propagation* in *Administration*.
    e. Select a stakeholder role.
    f. Click **Select** to add users and user groups from the stakeholder role as stakeholders for the technical assets.

       The **Add Users & User Groups** dialog box displays a list of users and user groups assigned to the selected stakeholder role.



g. Select one or more users or user groups to assign as stakeholders for the technical assets, and click **OK**.

   Only the selected users and user groups belonging to the specified stakeholder role are granted the permissions to technical assets.

h. To assign users or user groups from another stakeholder role, click **Add** and then repeat the steps.

2. To assign asset groups to technical assets extracted from the catalog source, perform the following steps:

   a. On the **Associations** page, click **Asset Groups**.

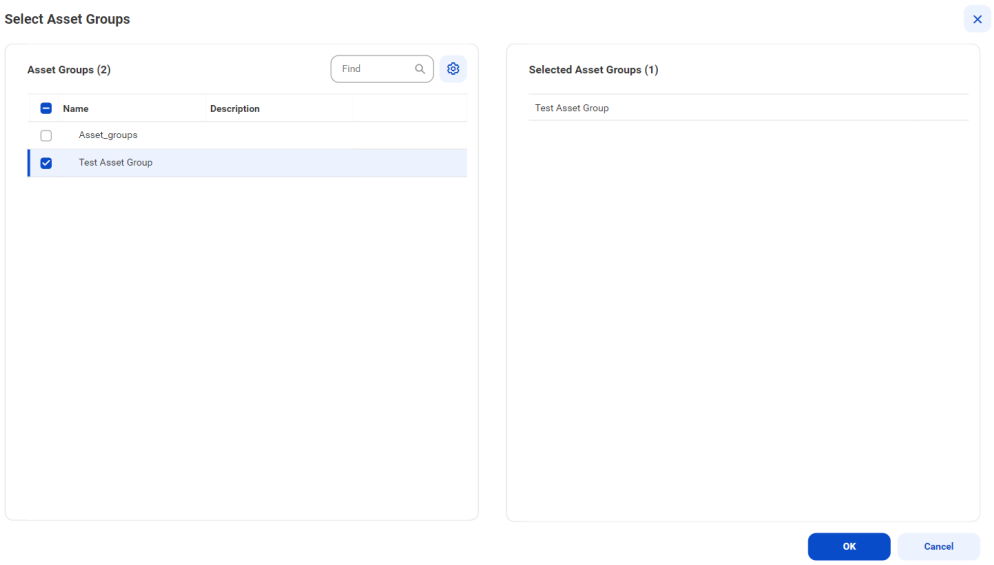
   b. Select **Assign Asset Groups**.

   c. Click **Select**.

      The **Select Asset Groups** dialog box displays the list of asset groups.

   If you enabled an access policy that includes an asset group, you can only view assets that belong to that asset group.

3. Select the asset groups to which you want to assign technical assets extracted from the catalog source, and click **OK**.



4. Choose to save and run the job or to schedule a recurring job.
   - To save and run the job, click **Save** and then **Run**.
   - To schedule a recurring job, click **Next** to open the **Schedule** page.

Step 4. Run or schedule the job

Choose to run a catalog source job manually, or configure it to run on schedule.

Note: You can't run multiple jobs simultaneously.

You can choose to perform a full or an incremental metadata extraction. A full metadata extraction extracts all objects from the source to the catalog. An incremental metadata extraction extracts only the changed and new objects since the last successful catalog source job run. Incremental metadata extraction doesn't remove deleted objects from the catalog and doesn't extract metadata of code-based objects if applicable.

When you run an incremental metadata extraction job with a filter to include metadata from objects, the job extracts only the objects that have the latest timestamp since the last successful job.

Note: If incremental metadata extraction is available for the catalog source, the option is always visible, but incremental metadata extraction jobs can run only after a scheduled full metadata extraction job. Manual jobs do not enable the incremental metadata extraction option.

Run the job manually

Click **Save** to save the catalog source and click **Run**. On the **Run Catalog Source Job** window, click **Run** to run the job.

You can override the capabilities that you selected while configuring your catalog source on the **Configuration** page. The first time you run the catalog source job, the metadata extraction capability is mandatory. From the second run onwards, you can choose to override the configured metadata change option. You can retain, delete, or deprecate objects that are deleted from the source in the catalog. For subsequent runs of the catalog source job, the metadata extraction capability is optional.

Note: You can choose incremental metadata extraction for subsequent runs only after one full metadata extraction job completes successfully. Incremental metadata extraction jobs run with the **Retain** metadata change option even if you set the option to **Delete** or **Deprecate** in the catalog source.

Note: To run a catalog source job, you need permissions on the connection to the source system. To run a catalog source job for catalog sources that reference other source systems, you need permissions on the connections for all the reference source systems.

Run the job on a schedule

You can choose to run metadata extraction and other capabilities on a recurring schedule. You can't choose incremental metadata extraction and full metadata extraction in the same schedule. To create a schedule for incremental metadata extraction, you must have completed at least one full metadata extraction job successfully. If not, first create a schedule for a full metadata extraction.

If an incremental metadata extraction is scheduled to run when the last run details aren't available, the job first performs a full metadata extraction, followed by incremental metadata extraction on subsequent runs.

For example, this can happen in the following scenarios:

- You create schedules for both incremental metadata extraction and full metadata extraction, but schedule the incremental extraction to run before the first full metadata extraction job.
- You create schedules for both incremental metadata extraction and full metadata extraction, but delete the full metadata extraction schedule before its first run.

1. On the **Schedule** tab, select **Run on Schedule**.
   The **Schedule** configuration page opens.
2. Click the checkbox corresponding to each capability that you want to include in the schedule.
3. Enter the start date, time zone, and the interval at which you want to run the job.
4. You can manage additional schedules using the following options:
   - To create a new schedule, click the **Add** button.
   - To delete a schedule, click the **Delete** button.
   - To enable or disable a schedule, click the **Enable Schedule** toggle button.

   Note: You can create a maximum of one schedule per capability that you enable. If you purged a catalog source or did not run the metadata extraction job, the catalog source job runs metadata extraction before running other scheduled capabilities.

   Note: To create a schedule, you need permissions on the connection to the source system. If you lose permissions on the connection after you create a schedule, the scheduled jobs continue to run.
5. Click **Save** to save the schedule.

Monitor job status

After the job runs, you can monitor the status of the job on the **Overview** page of the job.

For more information about job monitoring, see *Administration*.

CHAPTER 4

View results in Data Governance and Catalog

After Metadata Command Center runs a job, you can view the results in Data Governance and Catalog where the catalog source and its elements are called technical assets. You can view a catalog source as a hierarchy. Expand each technical asset to see its components.

When referenced source systems are connected to a catalog source, you can expand the hierarchy to see details about the technical asset's component elements.

You can view the data lineage of an asset contained within a catalog source to see individual elements such as data sources, calculations, and filters. When you view data lineage, you can see the individual upstream elements that contribute data or expressions to each component of a data flow or catalog source.

View metadata extraction results

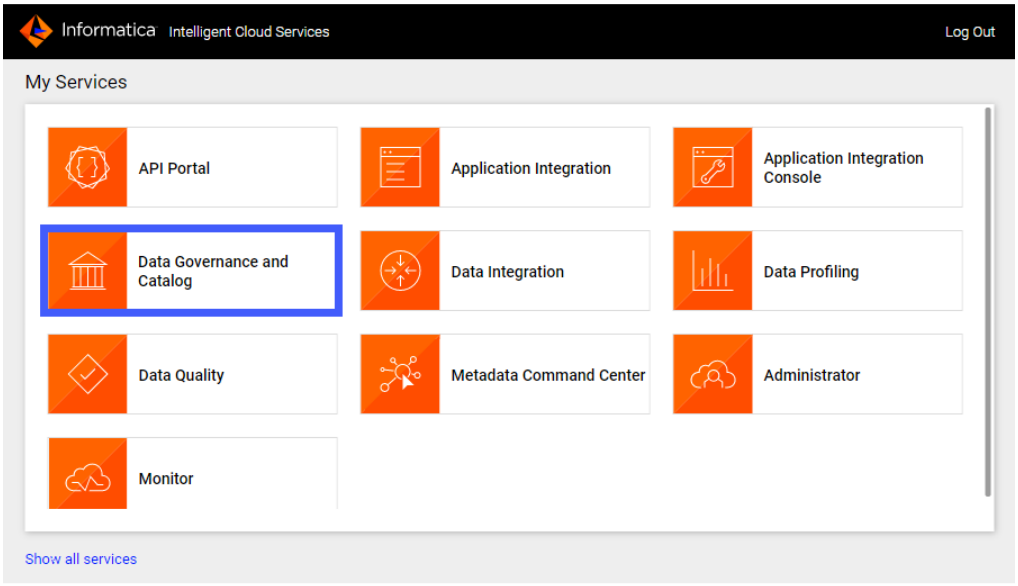
After a job runs in Metadata Command Center, view the results in Data Governance and Catalog. You can view details about source system contents in a hierarchical structure and trace data lineage.

1. Log in to Intelligent Data Management Cloud.

   The **My Services** page appears.

2. Click Data Governance and Catalog.

   The Data Governance and Catalog box appears on the **My Services** page.



3. On the Data Governance and Catalog home page, click the number in the **Technical Assets** panel.

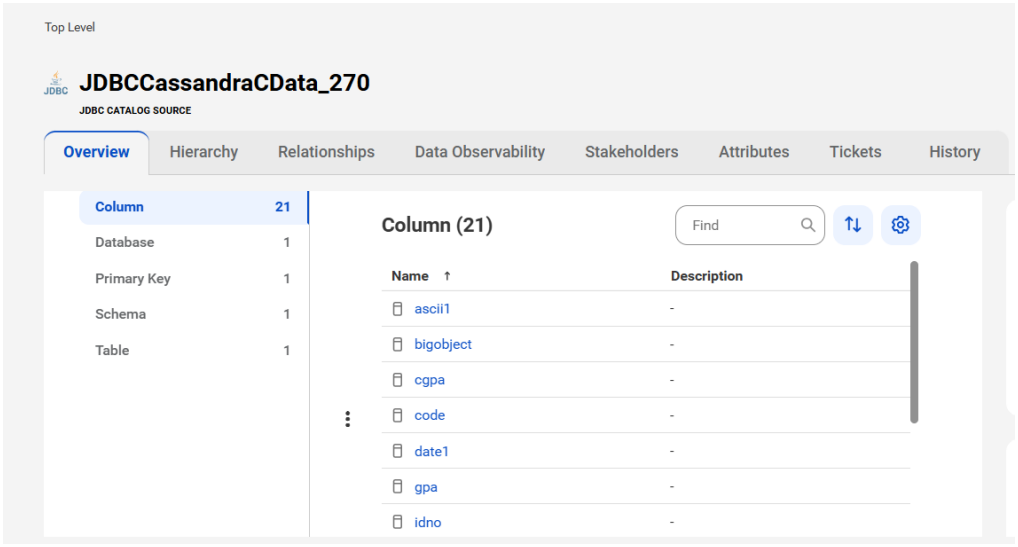
   The **Technical Assets** page opens.

4. Select **Catalog Source** in the **Filter** list.

   The list of catalog sources opens.

5. Search for the catalog source that you extracted metadata from, and click the name.

   The **Overview** tab of the asset opens.



6. View the asset from different perspectives by clicking on the tabs.

For more information about working with assets, see Working with Assets in *Data Governance and Catalog* help.

View relationships

Relationship views are available for technical assets in the catalog source. You can connect assets to each other using different types of relationships.

A relationship between assets shows how the assets relate to one another. When data from a source system is ingested into the catalog, Data Governance and Catalog can automatically create relationships among the technical assets of that source system.

For more information about viewing relationships, see *Relationships* in the Data Governance and Catalog help.

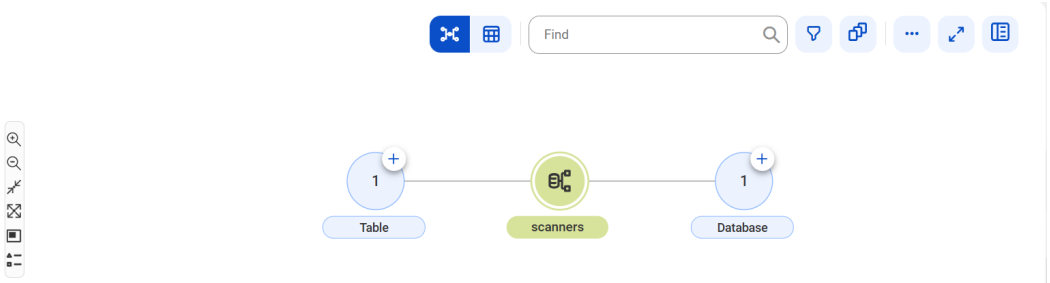
View relationships at catalog source level

The **Relationships** tab of an asset page displays the relationships between the selected asset and other business and technical assets in Data Governance and Catalog.

The JDBC catalog source reads metadata from the input data model and creates reference assets in the model.

To view relationships at the catalog source level, search for and open a JDBC catalog source, click the **Relationships** tab and expand the catalog source.

The following image shows the relationship between objects at the catalog source level:



View relationships at data set level and data element level

The relationships at the data set level and the data element level show how the technical assets such as tables and columns are related to the selected asset.

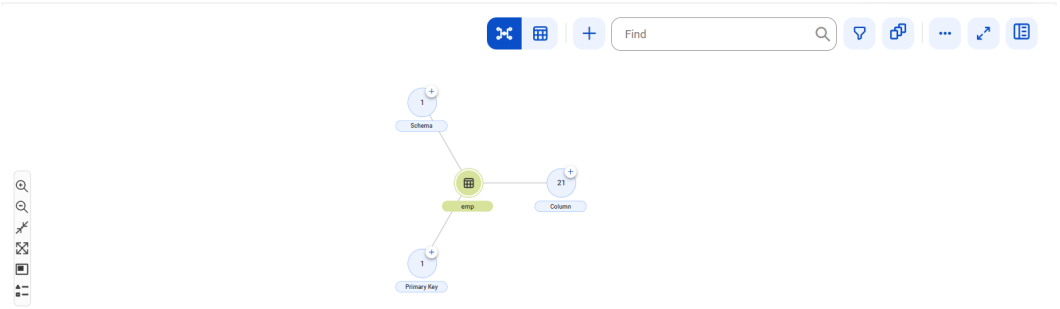
Data sets are technical assets that contain sets of data. For example, a table in a source object. Data elements are objects upstream or downstream of a data set. For example, a column in a source object.

View relationships at the data set level

The relationships at the data set level show how technical assets such as tables are related to the selected asset.

To view relationships at the data set level, open a technical asset such as a table, click the **Relationships** tab, and then expand the data set.

The following image shows the relationship at the emp table level:

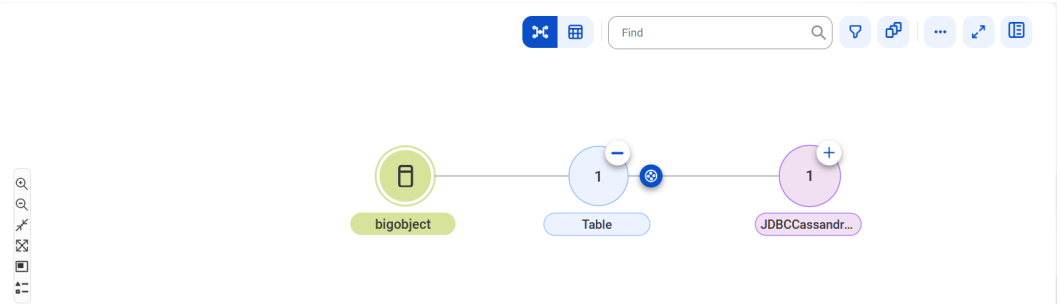


View relationships at the data element level

The data element level displays details of the data set level. The relationships at the data element level show how the technical assets such as columns are related to the selected asset.

To view relationships at the data element level, open a technical asset, click the **Relationships** tab, and then expand the data element.

The following image shows the relationship between objects at the data element level:

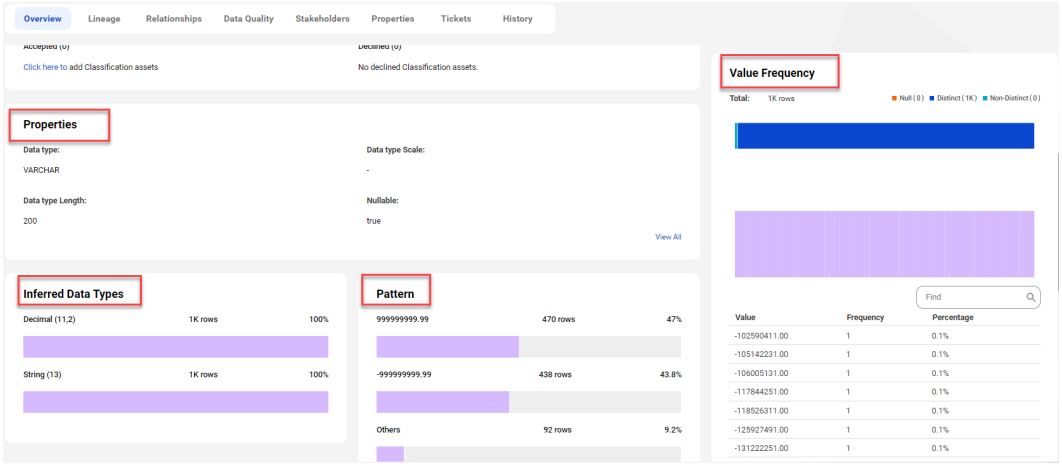


View data profiling results

When you enable the data profiling task for a catalog source in Metadata Command Center, the system runs a profile to evaluate the quality of the metadata extracted from the source system. The profiling statistics appear in Data Governance and Catalog when you open the technical assets.

The scope of profiling statistics that Data Governance and Catalog displays depends on the data profiling configuration parameters that you set when you configured the catalog source in Metadata Command Center.

The following image shows the data profiling statistics that appear on a column asset page in Data Governance and Catalog:



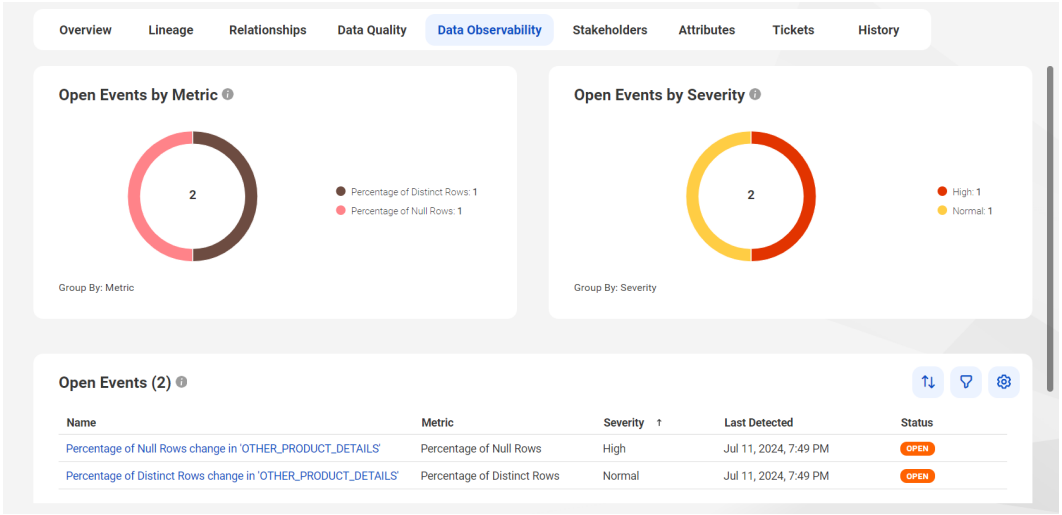
For more information about data profiling results, see *Asset Details* in the Data Governance and Catalog help.

View data observability results

When you enable data observability for a catalog source in Metadata Command Center, you can view and evaluate the events that it generates in Data Governance and Catalog. These events indicate anomalies identified in the characteristics of the profiled data in your source system.

You can view the events that data observability generates for anomalies identified for catalog sources, technical data sets, and data elements. You can then take appropriate actions for the generated events.
Note: The administrator of the catalog source might have applied filters to the data to narrow down the data elements that are applicable for business users in Data Governance and Catalog. The data for which users receive anomaly notifications depend on the filters that are configured for the catalog source.

The following image shows the open events for a column asset in Data Governance and Catalog:

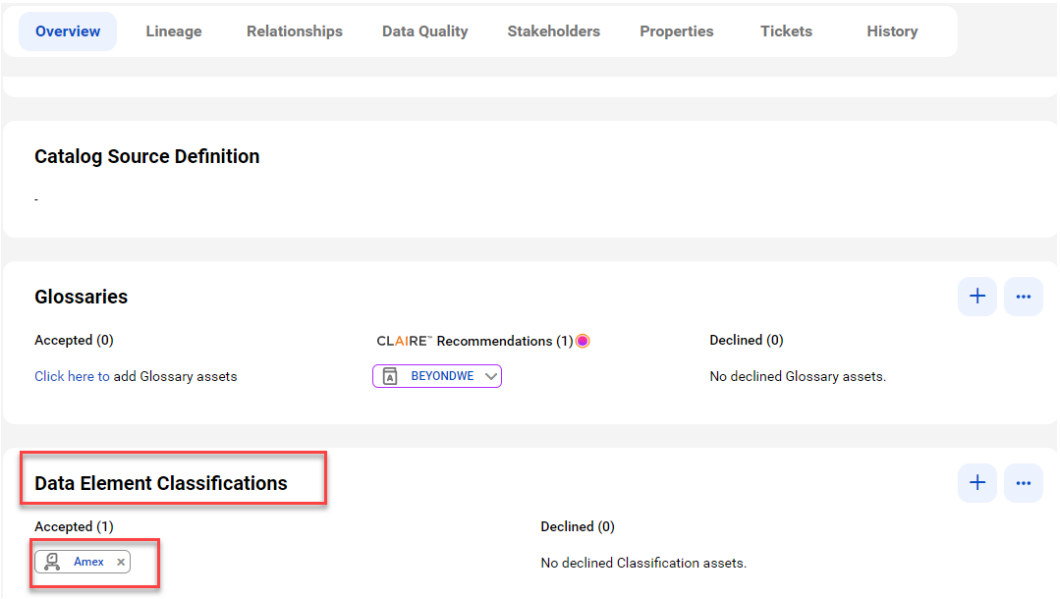


For more information about data observability results, see *Working With Assets* in the Data Governance and Catalog help.

View classified data

When you add data classification rules to a catalog source in Metadata Command Center, the system identifies the columns and tables that match the rules and displays one or more matched data classifications on the column or table asset pages in Data Governance and Catalog.

The following image shows a column asset page with the inferred data element classifications that match the column data and metadata:



For more information about data classification assets, see *Asset Details* in the Data Governance and Catalog help.

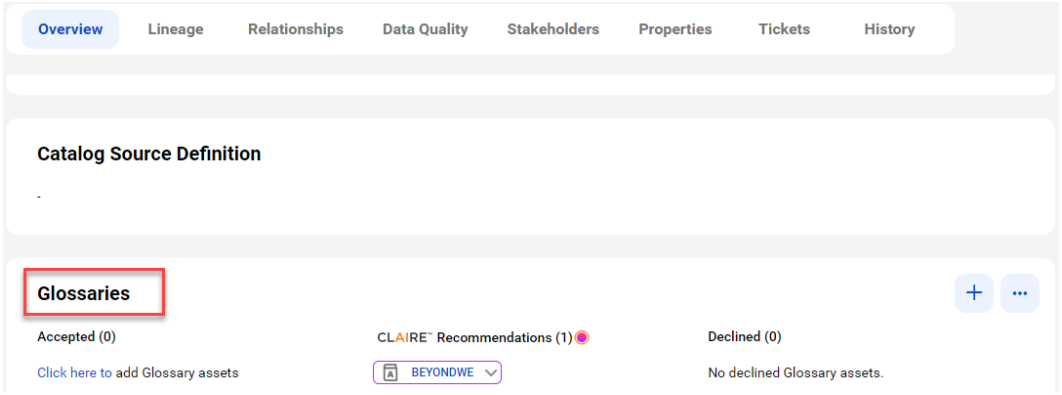
View glossary associations

When you enable the glossary association capability for a catalog source in Metadata Command Center, you can view the accepted glossary assets in Data Governance and Catalog.

The **Overview** tab for a technical asset in the catalog source displays glossary assets in the Accepted and CLAIRE Recommendations sections.

The **Glossaries** panel shows the automatically accepted and CLAIRE® recommended terms.

The following image shows a sample asset page:



APPENDIX A

Managing catalog sources

You can update, copy, delete, purge, and run a catalog source. You can also grant configuration permissions for catalog sources to different roles in your organization.

On the **Catalog Sources** page of the **Explore** page, you can select a catalog source to see its overview and related catalog sources. The **Overview** tab displays the capabilities, filters, stakeholders, asset groups, and schedules enabled for a catalog source. The **Related Catalog Sources** tab displays a list of related catalog sources that are either assigned manually using connection assignment or are curated based on CLAIRE recommendations using the lineage discovery capability. You can view related catalog sources and accept or reject CLAIRE recommendations for catalog sources that are run with the lineage discovery capability.

Cloning a catalog source configuration

You can clone the configuration of a catalog source to create another catalog source with the same configuration.

Note: Cloning doesn't copy the content of the catalog source.

1. In Metadata Command Center, go to the **Explore** page.
2. Select **Catalog Sources** from the menu.
3. You can clone a catalog source in one of the following ways:
   - Hover your mouse over the catalog source that you want to clone and click **Clone** from the **Action** menu.
   - Open the catalog source that you want to clone and click **Clone**.

   Note: By default, the cloned catalog source has the same name suffixed with _Clone. You can change the name if needed.
4. If needed, modify the configuration of the catalog source by updating the parameters on each tab.

   Cloning doesn't copy the content of sensitive fields. If the catalog source contains sensitive fields such as Client Secret, enter the values.
5. Click **Save**.

Deleting or purging a catalog source

Delete or purge a catalog source based on your business requirements.

When you delete a catalog source, you delete all the configuration specific to that catalog source, along with all the metadata that was extracted and any enrichment made to the associated metadata. Purging a catalog source deletes all the metadata that was extracted, along with any enrichment made to the associated metadata.

Note: Before you delete or purge a catalog source, manually unassign the connections that are assigned to that catalog source.

1. Go to the **Explore** page in Metadata Command Center.
2. Select **Catalog Sources** from the menu.
3. From the list of catalog sources, search for the catalog source that you want to delete or purge.
4. Click the **Action** menu, and select **Delete** or **Purge**.

   Note: You can't delete or purge a catalog source if the catalog source job is in the running state.
5. To confirm delete or purge, click **OK**.

   The catalog source is triggered to delete or purge. You can monitor the status of the job on the **Monitor** page.

   Note: If the deletion job succeeds, the catalog source disappears from Metadata Command Center. If it fails, some associated objects might still be present in Data Governance and Catalog. Such catalog sources continue to appear in the list of catalog sources in Metadata Command Center, but are marked as **Deleted**. Delete the catalog sources to remove them along with any remaining objects present in Data Governance and Catalog.

Purging obsolete objects from a catalog source

You can purge obsolete objects from a catalog source.

When you configure or run a catalog source, you can choose to deprecate objects that are deleted from a source system. If you delete objects from the source system or make changes to the filter, the objects imported into the catalog before the change move to the Obsolete lifecycle in the catalog. You can purge obsolete objects from a catalog source. This permanently deletes all objects with the Obsolete lifecycle status, along with associated enrichments, from the catalog.

1. Go to the **Explore** page in Metadata Command Center.
2. Select **Catalog Sources** from the menu.
3. From the list of catalog sources, search for the catalog source from which you want to purge obsolete objects.
4. Click the **Action** menu, and select **Purge Obsolete Objects**.

   A warning message appears on the page.
5. Click **Yes** to confirm.

   A confirmation message appears on the page and the purge job starts. You can monitor the status of the job on the **Monitor** page. If the job succeeds, all obsolete objects present in the catalog source disappear from the catalog.

   Note: If the job fails, some obsolete objects might remain in the catalog. To remove the remaining obsolete objects, run the purge job again.

Curating CLAIRE recommendations

You can view related catalog sources and accept or reject CLAIRE recommendations for a catalog source.

Applicable to catalog sources that generate lineage.

1. Go to the **Explore** page in Metadata Command Center.
2. Select **Catalog Sources** from the menu.
3. From the list of catalog sources, search for the catalog source for which you want to curate CLAIRE recommendations.
4. Hover your mouse over the catalog source and click **View Lineage Discovery** from the **Action** menu.

   Note: The **View Lineage Discovery** option is available for catalog sources that are run with the lineage discovery capability.

   The **Configure** page appears with the catalog source filtered on the **Lineage Discovery** tab of the **Lineage** tab.
   The **Related Catalog Sources** tab displays a list of related catalog sources.
5. Hover your mouse over a related catalog source and click **Accept** or **Reject** from the **Action** menu.

   For more information about lineage discovery and curating CLAIRE recommendations, see *Lineage discovery* in the *Administration* help.

Managing configuration permissions for catalog sources

As a service administrator, you can grant configuration permissions for catalog sources to different roles in your organization, and add specific users and user groups to those roles so that they can perform operational activities on a catalog source.

To manage permissions for catalog source configuration, verify that at least one role has the catalog source configuration **Set Permissions** functionality enabled for Metadata Command Center in Administrator. In Administrator, create the required roles, assign users and user groups to the roles, and specify appropriate permissions to the roles.

You can select roles and add specific users or user groups to those roles to grant the following levels of permission for catalog source configuration:

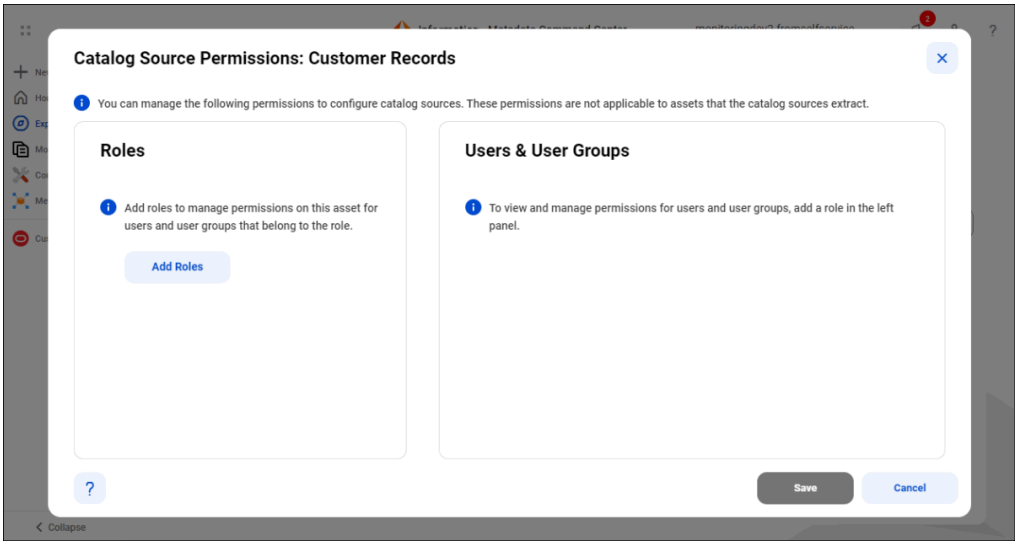
- Create
- Read
- Update
- Run
- Set Permissions

Based on the permissions granted to the roles, the selected users and user groups belonging to that role can perform appropriate operations for configuring catalog sources.

1. Go to the **Explore** page in Metadata Command Center.
2. From the list of catalog sources, search for the catalog source for which you want to manage permissions.

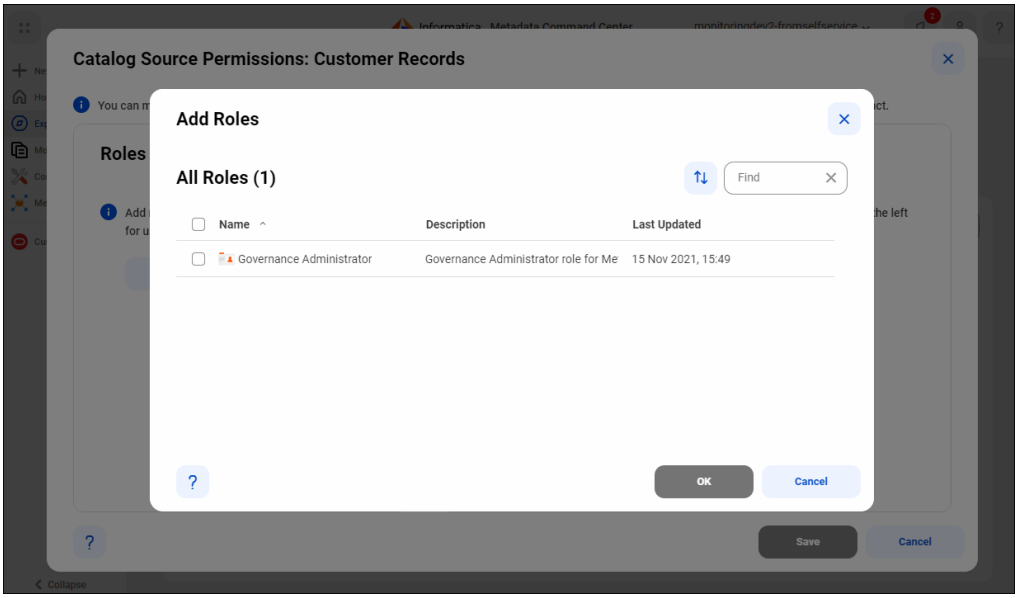
3. Click the **Action** menu, and select **Permissions**.

   The **Catalog Source Permissions** dialog box appears.



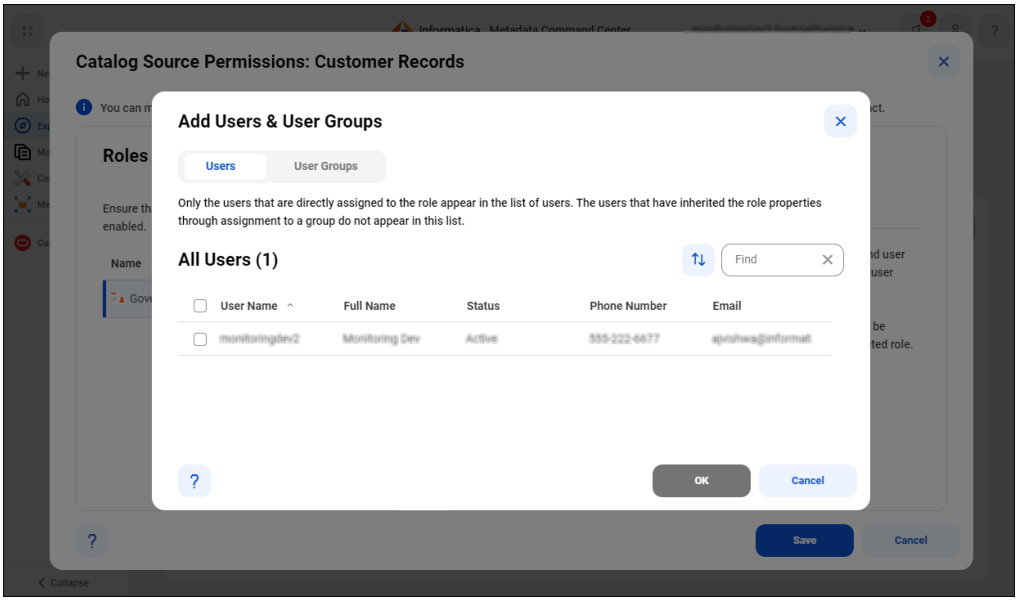
4. In the **Roles** panel, click **Add Roles**.

   The **Add Roles** dialog box appears.



5. Select one or more roles, and click **OK** to add the roles.
6. In the **Add Users & User Groups** panel, by default, all users and user groups that belong to the selected role are granted the role-defined permissions for the catalog source configuration.

To assign permissions only to the selected users and user groups that belong to this role, choose **Specific Users and User Groups**.



7. Select **Add Specific Users and User Groups**.

   The **Add Users & User Groups** dialog box displays the list of users and user groups that the administrator has assigned to the selected role.

8. Select one or more users or user groups to assign to the selected role, and click **OK**.

   Only the selected users and user groups belonging to the specified role will be granted the role-defined permissions for catalog source configuration.

9. View the assigned permissions for each user and or group, and click **Save**.